



powder and solvent for solution for injection

Active substance: Cetrorelix acetate

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains 0.25 mg cetrorelix (as acetate). After reconstitution with the solvent provided, each ml of the solution contains 0.25 mg cetrorelix.

For a full list of excipients, see List of Excipients.

PHARMACEUTICAL FORM

Powder and solvent for solution for injection.

Appearance of the powder: white lyophilisate.

Appearance of the solvent: clear and colourless solution.

The pH of the reconstituted solution is 4.0 – 6.0.

CLINICAL PARTICULARS

Therapeutic indications

Prevention of premature ovulation in patients undergoing a controlled ovarian stimulation, followed by oocyte pick-up and assisted reproductive techniques.

In clinical trials Cetrotide® 0.25 mg was used with human menopausal gonadotropin (HMG), however, limited experience with recombinant follicle-stimulating hormone (FSH) suggested similar efficacy.

Posology and method of administration

Cetrotide® 0.25 mg should only be prescribed by a specialist experienced in this field.

Posology

The first administration of Cetrotide® should be performed under the supervision of a physician and under conditions where treatment of possible allergic/pseudo-allergic reactions (including life-threatening anaphylaxis) is immediately available. The following injections may be self-administered as long as the patient is made aware of the signs and symptoms that may indicate hypersensitivity, the consequences of such a reaction and the need for immediate medical intervention.

The contents of 1 vial (0.25 mg cetrorelix) are to be administered once daily, at 24 h intervals, either in the morning or in the evening. Following the first administration, it is advised that the patient be kept under medical supervision for 30 minutes to ensure there is no allergic/pseudo-allergic reaction to the injection. Facilities for the treatment of such reactions should be immediately available.

Elderly

There is no relevant use of Cetrotide in the geriatric population.

Paediatric population

There is no relevant use of Cetrotide in the paediatric population.

Method of administration

Cetrotide is for subcutaneous injection into the lower abdominal wall.

The injection site reactions may be minimised by rotating the injection sites, delaying injection at the same site and injecting the medicinal product in a slow rate to facilitate the progressive absorption of the medicinal product.

Administration in the morning: Treatment with Cetrotide® 0.25 mg should commence on day 5 or 6 of ovarian stimulation (approximately 96 to 120 hours after start of ovarian stimulation) with urinary or recombinant gonadotropins and is to be continued throughout the gonadotropin treatment period including the day of ovulation induction.

The starting day of Cetrotide is depending on the ovarian response, i.e. the number and size of growing follicles and/or the amount of circulating oestradiol. The start of Cetrotide may be delayed in absence of follicular growth, although clinical experience is based on starting Cetrotide on day 5 or day 6 of stimulation.

Administration in the evening: Treatment with Cetrotide® 0.25 mg should commence on day 5 of ovarian stimulation (approximately 96 to 108 hours after start of ovarian stimulation) with urinary or recombinant gonadotropins and is to be continued throughout the gonadotropin treatment period until the evening prior to the day of ovulation induction.

The starting day of Cetrotide is depending on the ovarian response, i.e. the number and size of growing follicles and/or the amount of circulating oestradiol. The start of Cetrotide may be delayed in absence of follicular growth, although clinical experience is based on starting Cetrotide on day 5 or day 6 of stimulation.

Contraindications

- Hypersensitivity to the active substance or any structural analogues of gonadotropin-releasing hormone (GnRH), extrinsic peptide hormones or to any of the excipients.
- Pregnancy and lactation.
- Patients with severe renal impairment.

Special warnings and precautions for use

Allergic reactions

Cases of allergic/pseudoallergic reactions, including life-threatening anaphylaxis with the first dose have been reported (see under Undesirable effects).

Special care should be taken in women with signs and symptoms of active allergic conditions or known history of allergic predisposition. Treatment with Cetrotide is not advised in women with severe allergic conditions.

Ovarian Hyperstimulation Syndrome (OHSS)

During or following ovarian stimulation an ovarian hyperstimulation syndrome can occur. This event must be considered as an intrinsic risk of the stimulation procedure with gonadotropins.

An ovarian hyperstimulation syndrome should be treated symptomatically, e.g. with rest, intravenous electrolytes/colloids and heparin therapy.

Luteal phase support should be given according to the reproductive medical centre's practice.

Repeated ovarian stimulation procedure

There is limited experience up to now with the administration of Cetrotide during a repeated

ovarian stimulation procedure. Therefore Cetrotide should be used in repeated cycles only after a careful risk/benefit evaluation.

Congenital anomalies

The prevalence of congenital anomalies after the use of assisted reproductive technologies (ART) with or without GnRH antagonists may be slightly higher than after spontaneous conceptions although it is unclear whether this is related to factors inherent to the couple's infertility or the ART procedures. Limited data from clinical follow-up studies in 316 newborns of women administered cetrorelix for infertility treatments suggest that cetrorelix does not increase the risk of congenital anomalies in the offsprings.

Hepatic impairment

Cetrorelix has not been studied in patients with hepatic impairment and caution is therefore warranted.

Renal impairment

Cetrorelix has not been studied in patients with renal impairment and caution is therefore warranted.

Cetrorelix is contraindicated in patients with severe renal impairment (see under Contraindications).

Interaction with other medicinal products and other forms of interaction

No formal drug-drug interaction studies have been performed with cetrorelix. In vitro investigations have shown that interactions are unlikely with medicinal products that are metabolised by cytochrome P450 or glucuronised or conjugated in some other way. However, the possibility of interactions with gonadotropins or medicinal products that may induce histamine release in susceptible individuals, cannot be totally excluded.

Fertility, Pregnancy and lactation

Pregnancy and Breast-feeding

Cetrotide is not intended to be used during pregnancy and lactation (see Contraindications).

Fertility

Studies in animals have indicated that cetrorelix exerts a dose related influence on fertility, reproductive performance and pregnancy. No teratogenic effects occurred when the medicinal product was administered during the sensitive phase of gestation.

Effects on ability to drive and use machines

Cetrotide has no or negligible influence on the ability to drive or use machines.

Undesirable effects

Summary of the safety profile

The most commonly reported side effects are local injection site reactions such as erythema, swelling and pruritus that are usually transient in nature and mild in intensity. In clinical trials, these effects were observed with a frequency of 9.4% following multiple injections of Cetrotide® 0.25 mg.

Mild to moderate ovarian hyperstimulation syndrome (OHSS) (WHO grade I or II) have been commonly reported and should be considered as an intrinsic risk of the stimulation procedure. Inversely, severe OHSS remains uncommon.

Uncommonly, cases of hypersensitivity reactions including pseudo-allergic/anaphylactoid reactions have been reported.

List of adverse reactions

The adverse reactions reported below are classified according to frequency of occurrence as follows: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$), very rare ($< 1/10,000$).

Immune system disorders

Uncommon: Systemic allergic/pseudo-allergic reactions including life-threatening anaphylaxis.

Nervous system disorders

Uncommon: Headache

Gastrointestinal disorders

Uncommon: Nausea

Reproductive system and breast disorders

Common: Mild to moderate OHSS (WHO grade I or II) can occur which is an intrinsic risk of the stimulation procedure (see under Special warnings and precautions for use).

Uncommon: Severe OHSS (WHO grade III)

General disorders and administration site conditions

Common: Local reactions at the injection site (e.g. erythema, swelling and pruritus).

Overdose

Overdosage in humans may result in a prolonged duration of action but is unlikely to be associated with acute toxic effects.

In acute toxicity studies in rodents non-specific toxic symptoms were observed after intraperitoneal administration of cetrorelix doses more than 200 times higher than the pharmacologically effective dose after subcutaneous administration.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: anti-gonadotropin-releasing hormones, ATC code: H01CC02.

Mechanism of action

Cetrorelix is a luteinising hormone releasing hormone (LHRH) antagonist. LHRH binds to membrane receptors on pituitary cells. Cetrorelix competes with the binding of endogenous LHRH to these receptors. Due to this mode of action, cetrorelix controls the secretion of gonadotropins (LH and FSH).

Cetrorelix dose-dependently inhibits the secretion of LH and FSH from the pituitary gland. The onset of suppression is virtually immediate and is maintained by continuous treatment, without initial stimulatory effect.

Clinical efficacy and safety

In females, cetrorelix delays the LH surge and consequently ovulation. In women undergoing ovarian stimulation the duration of action of cetrorelix is dose dependent. At a dose of 0.25 mg per injection repeated injections every 24 hours will maintain the effect of cetrorelix.

In animals as well as in humans, the antagonistic hormonal effects of cetrorelix were fully reversible after termination of treatment.

Pharmacokinetic properties

Absorption

The absolute bioavailability of cetrorelix after subcutaneous administration is about 85%.

Distribution

The volume of distribution (Vd) is $1.1 \text{ L} \times \text{kg}^{-1}$.

Elimination

The total plasma clearance and the renal clearance are $1.2 \text{ ml} \times \text{min}^{-1} \times \text{kg}^{-1}$ and $0.1 \text{ ml} \times \text{min}^{-1} \times \text{kg}^{-1}$, respectively.

The mean terminal half-lives following intravenous and subcutaneous administration are about 12 h and 30 h, respectively, demonstrating the effect of absorption processes at the injection site.

Linearity

The subcutaneous administration of single doses (0.25 mg to 3 mg cetorelix) and also daily dosing over 14 days show linear kinetics.

Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, toxicity to reproduction.

No target organ toxicity could be observed from acute, subacute and chronic toxicity studies in rats and dogs following subcutaneous administration of cetorelix. No signs of medicinal product-related local irritation or incompatibility were noted in dogs after intravenous, intra-arterial and paravenous injection when cetorelix was administered in doses clearly above the intended clinical use in man.

Cetorelix showed no mutagenic or clastogenic potential in gene and chromosome mutation assays.

PHARMACEUTICAL PARTICULARS

List of excipients

Powder:

Mannitol

Solvent:

Water for injections

Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in Special precautions for disposal and other handling.

Shelf life

24 months

Special precautions for storage

Store in a refrigerator ($2^{\circ}\text{C} - 8^{\circ}\text{C}$). Do not freeze or place next to the freezer compartment or a freezer pack.

Store in the original package in order to protect from light.

The patient may remove the product from the refrigerator and store it not above 30°C for one single period of up to a maximum of 3 months.

This medicinal product must be allowed to reach room temperature prior to injection. It should be removed from the refrigerator approximately 30 minutes before use.

Nature and contents of container

Powder

2 ml vials (Type I glass) with a stopper (bromobutyl rubber) and flip-off aluminium cap.
1 vial contains 0.25 mg cetorelix.

Solvent

Pre-filled syringe (Type I glass) with plunger stopper (siliconised bromobutyl rubber) and tip cap (polypropylene and styrene butadiene rubber).
1 pre-filled syringe contains 1 ml of water for injections.

Pack sizes

1 vial and 1 pre-filled syringe or 7 vials and 7 pre-filled syringes.
Additionally, for each vial the pack contains:

1 injection needle (20 gauge)
1 hypodermic injection needle (27 gauge)
2 alcohol swabs

Not all pack sizes may be marketed.

Special precautions for disposal and other handling

This medicinal product must be allowed to reach room temperature prior to injection. It should be removed from the refrigerator approximately 30 minutes before use.

Cetrotide should only be reconstituted with the solvent provided, using a gentle, swirling motion. Vigorous shaking with bubble formation should be avoided

The reconstituted solution is without particles and clear. Do not use if the solution contains particles or if the solution is not clear

The entire contents of the vial should be withdrawn to ensure a delivery to the patient of a dose of at least 0.23 mg cetorelix.

The solution should be used immediately after reconstitution.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

How Should Cetrotide ® 0.25 mg be used

This section tells you how to mix the powder and the sterile water (solvent) together and then how to inject your medicine.

You may self-administer Cetrotide ® 0.25 mg after appropriate instruction from your doctor.

This medicine is only for you – do not let anyone else use it.

Cetrotide ® 0.25 mg is for injection under the skin of the lower abdominal wall, preferably around the navel. To minimize local irritation, please select a different injection site each day.

Dissolve Cetrotide ® 0.25 mg powder only with the water contained in the pre-filled syringe. Do not use a Cetrotide ® 0.25 mg solution if it contains particles or if it is not clear.

Before starting to use this medicine, please read these instructions the whole way through first.

1. This medicine must be at room temperature prior to injection. Remove from the refrigerator approximately 30 minutes before use.

2. Wash your hands

- It is important that your hands and the things you use are as clean as possible.

3. Lay out everything you need on a clean surface:

- one vial of powder
- one pre-filled syringe with sterile water (solvent)
- one needle with a **yellow** mark – for injecting the sterile water into the vial and drawing the made-up medicine out from the vial
- one needle with a **grey** mark – for injecting the medicine into your belly
- two alcohol swabs

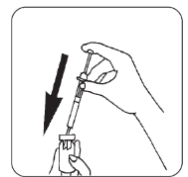
Mixing the powder and water to make up your medicine

1. Remove the cap from the vial

- There will be a rubber stopper underneath – keep this in the vial.
- Wipe the rubber stopper and metal ring with your first alcohol swab.

2. Adding the water from the pre-filled syringe to the powder in the vial

- Unwrap the needle with the **yellow** mark on it.
- Remove the cap from the pre-filled syringe and screw the yellow needle onto it. Remove the cap from the needle.
- Push the yellow needle through the centre of the rubber stopper of the vial.
- Slowly push in the plunger of the syringe to inject the water into the vial. Do not use any other sort of water.
- Leave the syringe in the rubber stopper.

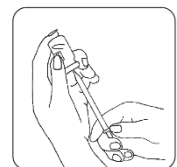


3. Mixing the powder and water in the vial

- While carefully holding the syringe and vial, swirl gently to mix the powder and water together. When it is mixed, it will look clear and have no particles in it.
- Do not shake or you will create bubbles in your medicine.

4. Re-filling the syringe with the medicine from the vial

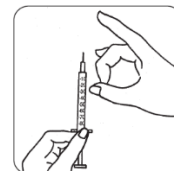
- Turn the vial upside down. Then, gently pull back the plunger, in order to draw the medicine from the vial into the syringe. Take care not to pull out completely the plunger with the attached plunger stopper. In case you pulled out the plunger with the plunger stopper by mistake, make sure to discard the dose as sterility will be lost, and prepare a new dose (and restart from step 1).
- If any medicine is left in the vial, pull out the yellow needle until the end of the needle is just inside the rubber stopper. If you look from the side through the gap in the rubber stopper, you can control the movement of the needle and the liquid.
- Make sure that you collect all of your medicine from the vial.
- Put the cap back on the yellow needle. Unscrew the yellow needle from the syringe and lay down the syringe.



Preparing the injection site and injecting your medicine

1. Removing air bubbles

- Unwrap the needle with the **grey** mark on it. Screw the grey needle onto the syringe and remove the cap from the grey needle.
- Hold the syringe with the grey needle pointing upwards and check for any air bubbles.
- To remove air bubbles, gently flick the syringe until all the air collects at the top – then slowly push the plunger in until the air bubbles are gone.
- Do not touch the grey needle and do not let the needle touch any surface.

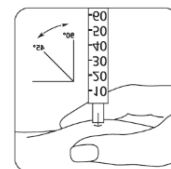


2. Clean the injection site

- Choose an injection site on your belly. It is best around the belly button (navel). To reduce skin irritation, select a different part of your belly each day.
- Clean the skin at your chosen injection site with your second alcohol swab - use a circular motion.

3. Piercing your skin

- Hold the syringe in one hand – like you would hold a pencil.
- Gently pinch up the skin around where you are going to inject and hold this firmly with your other hand.
- Slowly push the grey needle completely into your skin at an angle of about 45 to 90 degrees – then let go of your skin.



4. Injecting your medicine

- Gently pull back the plunger of the syringe. If blood appears, follow Step 5 below.
- If no blood appears, **slowly** push the plunger in to inject your medicine.
- When the syringe is empty, take out the grey needle slowly at the same angle.
- Use your second alcohol swab to gently apply pressure where you have just injected.

5. If blood appears:

- take out the grey needle slowly at the same angle
- use your second alcohol swab to gently apply pressure where you have just pierced your skin
- empty your medicine into a sink and follow Step 6 below
- wash your hands and start again with a new vial and pre-filled syringe.

6. Disposal

- Use each needle, vial and syringe only once.
- Put the cap back on the needles so that they are safe to be thrown away.
- Ask your pharmacist how to safely dispose of used needles, vial and syringe.

MANUFACTURER

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